

2024, Mar. 2024 Release LOC:Acute Adult

Asthma

Utilization Benchmarks: Multiple options available, see table below for options

Overview

Select Day

Initial review (1, 2, 3, 4)

Episode Day 1 (1, 3, 4)

Episode Day 2 (4, 37, 38)

Episode Day 3 (4, 38, 48)

Episode Day 4

Discharge Screens (49)

Utilization Benchmarks

Notes

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Overview

Introduction:

Asthma is a chronic inflammatory disorder of both the large and small airways causing recurrent episodes of wheezing, breathlessness, chest tightness, and coughing. These episodes are usually associated with widespread but variable airflow obstruction and may reverse spontaneously or with treatment.

Asthma exacerbations are characterized by decreases in expiratory airflow that can be quantified and documented by measurement peak expiratory flow (PEF) or forced expiratory volume in 1 second (FEV₁). These measurements, when obtainable, can more accurately reflect the severity of the exacerbation than the symptoms being exhibited.

Evaluation and Treatment:

The decision regarding appropriate disposition is based on duration and severity of symptoms, severity of airflow obstruction, response to initial treatment and risk factors for poor outcomes.

The initial assessment includes a brief history and physical examination, assessment of PEF or FEV₁, determination of oxygen saturation, and other tests as indicated. Treatment for acute asthma exacerbation may include (Global Initiative for Asthma (GINA), Global Strategy for Asthma Management and Prevention. 2023; National Heart, Lung, and Blood Institute. Expert Panel Report 3: Guidelines for the Diagnosis and Management of Asthma - Full Report 2007):

- Inhaled short-acting bronchodilators (SABA) and ipratropium
- Systemic corticosteroids
- Supplemental oxygen, when necessary, to maintain oxygen saturation at 93-95%(0.93-0.95)
- Impending or actual respiratory arrest: noninvasive positive pressure ventilation, intubation, and mechanical ventilation
- Magnesium sulfate and heliox (mixture of helium and oxygen) may be administered as adjunctive therapy

After emergency department (ED) treatment, many patients with asthma may be cared for in the observation unit while waiting for treatments given in the ED to take full effect. Patients with asthma can be monitored in the observation unit to assess treatment effect and also to allow time for patient education which may decrease future ED visits or hospitalizations. It is estimated that at least one-third of United States hospitals have a dedicated observation unit. However, observation units are underutilized as evidenced by the number of inpatient admissions for asthma short-stay admissions (Pena et al., Emerg Med Clin North Am 2017, 35: 625-45; Napolitano, Observation Units: Definition, History, Data, Financial Considerations, and Metrics. 2014).

InterQual® criteria are derived from the systematic, continuous review and critical appraisal of the most current evidence-based literature and include input from our independent panel of clinical experts. To generate the most

appropriate recommendations, a comprehensive literature review of the clinical evidence was conducted. Sources searched included PubMed, AHRQ Comparative Effectiveness Reviews, the National Institute for Health and Care Excellence (NICE), the Centers for Medicare and Medicaid Services (CMS), The Cochrane Library, and The Joint Commission. Other medical literature databases, medical content providers, data sources, regulatory body websites, and specialty society resources may also have been utilized. Relevant studies were assessed for risk of bias following principles described in the *Cochrane Handbook*. The resulting evidence was assessed for consistency, directness, precision, effect size, and publication bias. Observational trials were also evaluated for the presence of a dose-response gradient and the likely effect of plausible confounders. The majority of the content in this subset is based on a limited number of key sources (Global Initiative for Asthma (GINA), Global Strategy for Asthma Management and Prevention. 2023; National Heart, Lung, and Blood Institute. Expert Panel Report 3: Guidelines for the Diagnosis and Management of Asthma - Full Report 2007).

In an effort to address inequities in medicine and improve appropriate access to care, multiple medical societies have been reevaluating the use of adjustments made for a patient's race and updating their calculators and reference equations when reporting diagnostic results.

In 2023, the American Thoracic Society issued updated guidance on reporting of pulmonary function tests (PFTs), advocating for a race-neutral average reference equation to determine the baseline (i.e., predicted normal) for a given patient (Bhakta et al., Am J Respir Crit Care Med 2023, 207: 978-995). The reference equation previously included lower predicted normal values for patients of certain races and ethnicities (in the US, for Black and Asian patients); a practice which has not been found to have biological validity and may lead to misclassification of diagnoses and disease severity in racial or ethnic minorities (Vyas et al., N Engl J Med 2020, 383: 874-882).

Since not all spirometer manufacturers and PFT laboratories have switched over to the new race-neutral average reference equation, clinical judgement and consideration of patient-specific factors should be used when interpreting PFT results and deciding whether patients with borderline reported PFT values in the presence of concerning symptoms or signs should have further evaluation or treatment.

(Excludes PO medications unless noted)

Select Day, One:● **Initial review, One:** ^(1, 2, 3, 4)

(From arrival in ED through decision to admit)

(Symptom or finding within 24h)

● **OBSERVATION, One:** ⁽⁵⁾● High risk for decompensation, ≥ **One:** ⁽⁶⁾

- Comorbidity ^(7, 8)
- History of asthma requiring intubation and mechanical ventilation
- Hospitalization or ED visit for asthma within the past year ⁽⁹⁾
- Oral steroid use, current or within the past month ⁽¹⁰⁾
- Poor adherence with asthma medications and/or poor adherence with an asthma action plan ⁽¹¹⁾
- Pregnancy
- Use of > 1 canister of short-acting beta-agonist per month

● Partial response to at least 2h of initial treatment and, **One:** ^(12, 13, 14, 15, 16)● Wheezing, persistent and, ≥ **One:** ⁽¹⁷⁾

- O₂ sat < 95%(0.95) and < baseline ^(18, 19)
- PEF or forced expiratory volume at one second (FEV₁) 40-69%(0.40-0.69) or symptoms of airway obstruction ^(20, 21, 22, 23)
- Respiratory rate 22-30/min, sustained ^(24, 25)

● **CRITICAL, One:** ⁽²⁶⁾● Impending respiratory failure, ≥ **One:** ⁽²⁷⁾

- Status asthmaticus ⁽²⁸⁾
- Arterial Pco₂ ≥ 42 mmHg(5.6 kPa) ⁽²⁹⁾
- Confusion or drowsiness

● Poor response after at least 2h of initial treatment and, ≥ **One:** ^(12, 13, 14, 16)

- Accessory muscle use ⁽³⁰⁾
- PEF or forced expiratory volume at one second (FEV₁) < 40%(0.40) ^(20, 21, 22, 23, 31)

● **Episode Day 1, One:** ^(1, 3, 4)

(Symptom or finding within 24h)

● **OBSERVATION, One:** ⁽⁵⁾

- Pneumonia (**use Infection: Pneumonia subset**)
- Acute heart failure (**use Heart Failure subset**)
- Suspected ACS (**use Acute Coronary Syndrome subset (ACS)**)

● High risk for decompensation, **Both:** ⁽⁶⁾● Risk factor, ≥ **One:**

- Comorbidity ^(7, 8)
- History of asthma requiring intubation and mechanical ventilation
- Hospitalization or ED visit for asthma within the past year ⁽⁹⁾
- Oral steroid use, current or within the past month ⁽¹⁰⁾
- Poor adherence with asthma medications and/or poor adherence with an asthma action plan ⁽¹¹⁾
- Pregnancy
- Use of > 1 canister of short-acting beta-agonist per month

● Intervention, **Both:**

- Corticosteroid (includes PO) ⁽³²⁾
- Short-acting beta-agonist ≥ 4x/24h ⁽³³⁾

● Partial response to at least 2h of initial treatment and, **Both:** ^(12, 13, 14, 15, 16)● Wheezing, persistent and, ≥ **One:** ⁽¹⁷⁾

- O₂ sat < 95%(0.95) and < baseline ^(18, 19)
- PEF or forced expiratory volume at one second (FEV₁) 40-69%(0.40-0.69) or symptoms of airway obstruction ^(20, 21, 22, 23)
- Respiratory rate 22-30/min, sustained ^(24, 25)

● Intervention, **Both:**

- Corticosteroid (includes PO) ^(12, 32)
- Short-acting beta-agonist $\geq 4x/24h$ ⁽³³⁾
- **CRITICAL, Both:** ⁽²⁶⁾
 - Impending respiratory failure, $\geq$ **One:** ⁽²⁷⁾
 - Status asthmaticus ⁽²⁸⁾
 - Arterial $P_{CO_2} \geq 42$ mmHg(5.6 kPa) ⁽²⁹⁾
 - Confusion or drowsiness
 - Poor response after at least 2h of initial treatment and, $\geq$ **One:** ^(12, 13, 14, 16)
 - Accessory muscle use ⁽³⁰⁾
 - PEF or forced expiratory volume at one second (FEV_1) $< 40\%$ (0.40) ^(20, 21, 22, 23, 31)
 - Intervention, $\geq$ **One:**
 - Heliox ⁽³⁴⁾
 - Mechanical ventilation
 - NIPPV ^(35, 36)
 - Short-acting beta-agonist every 1-2h or continuous ⁽³³⁾
- **Episode Day 2, One:** ^(4, 37, 38)
 - **OBSERVATION, One:** ⁽⁵⁾
 - **Responder**, discharge expected today if clinically stable last 12h, **All:** ⁽³⁹⁾
 - Wheezing, resolved or resolving
 - O_2 sat within acceptable limits ⁽⁴⁰⁾
 - PEF or forced expiratory volume at one second (FEV_1) within acceptable limits or no symptoms of airway obstruction ^(22, 40, 41)
 - Vital signs stable or within acceptable limits ⁽⁴⁰⁾
 - No complication or active comorbidity relevant to this episode of care ⁽⁴²⁾
 - **Partial responder**, not clinically stable for discharge and requires continued stay, **Both:**
 - Wheezing, persistent and, $\geq$ **One:**
 - O_2 sat $< 95\%$ (0.95) and $<$ baseline ⁽¹⁸⁾
 - PEF or forced expiratory volume at one second (FEV_1) 40-69%(0.40-0.69) or symptoms of airway obstruction ^(20, 21, 22, 23)
 - Respiratory rate 22-30/min, sustained ^(24, 25)
 - Intervention, **Both:**
 - Corticosteroid (includes PO) ⁽³²⁾
 - Short-acting beta-agonist $\geq 4x/24h$ ⁽³³⁾
 - **ACUTE, One:** ⁽⁴³⁾
 - Post Critical care $\leq 24h$
 - **INTERMEDIATE, $\geq$ One:** ⁽⁴⁴⁾
 - NIPPV ^(35, 36)
 - Oxygen $\geq 40\%$ (0.40) $\leq 2d$ ^(45, 46)
 - **CRITICAL, $\geq$ One:** ⁽²⁶⁾
 - Heliox ⁽³⁴⁾
 - Mechanical ventilation
 - NIPPV and, $\geq$ **One:** ^(35, 36)
 - $FiO_2 > 50\%$ (0.50)
 - Respiratory intervention every 1-2h ⁽⁴⁷⁾
 - Short-acting beta-agonist every 1 - 2h or continuous ⁽³³⁾
 - **Episode Day 3, One:** ^(4, 38, 48)
 - **OBSERVATION, One:** ⁽⁵⁾
 - **Responder**, discharge expected today if clinically stable last 12h, **All:** ⁽³⁹⁾
 - Wheezing, resolved or resolving
 - O_2 sat within acceptable limits ⁽⁴⁰⁾
 - PEF or forced expiratory volume at one second (FEV_1) within acceptable limits or no symptoms of airway obstruction ^(22, 40, 41)
 - Vital signs stable or within acceptable limits ⁽⁴⁰⁾

- No complication or active comorbidity relevant to this episode of care ⁽⁴²⁾
- **Non-responder**, not clinically stable for discharge and exceeds episode days at the Observation level of care for this condition, **One**:
 - For asthma use Episode Day 3 at a higher level of care
 - For a condition other than asthma use appropriate subset (Episode Day 1) based on clinical findings
 - Refer for secondary review
- **ACUTE, One**: ⁽⁴³⁾
 - **Responder**, discharge expected today if clinically stable last 12h, **All**: ⁽³⁹⁾
 - Wheezing, resolved or resolving
 - O₂ sat within acceptable limits ⁽⁴⁰⁾
 - PEF or forced expiratory volume at one second (FEV₁) within acceptable limits or no symptoms of airway obstruction ^(22, 40, 41)
 - Vital signs stable or within acceptable limits ⁽⁴⁰⁾
 - No complication or active comorbidity relevant to this episode of care ⁽⁴²⁾
 - **Partial responder**, not clinically stable for discharge and requires continued stay, ≥ **One**:
 - Wheezing, persistent and, **Both**:
 - Finding, ≥ **One**:
 - O₂ sat < 90%(0.90) and < baseline ⁽¹⁸⁾
 - PEF or forced expiratory volume at one second (FEV₁) < 70%(0.70) or symptoms of airway obstruction ^(20, 21, 22, 23)
 - Respiratory rate 22-30/min, sustained ^(24, 25)
 - Corticosteroid (includes PO) and, ≥ **One**: ⁽³²⁾
 - Short-acting beta-agonist ≥ 4x/24h ⁽³³⁾
 - Supplemental oxygen ^(45, 46)
 - Post Critical care ≤ 24h
 - **INTERMEDIATE, ≥ One**: ⁽⁴⁴⁾
 - NIPPV ^(35, 36)
 - Oxygen ≥ 40%(0.40) ≤ 2d ⁽⁴⁵⁾
 - **CRITICAL, ≥ One**: ⁽²⁶⁾
 - Heliox ⁽³⁴⁾
 - Mechanical ventilation
 - NIPPV and, ≥ **One**: ^(35, 36)
 - FiO₂ > 50%(0.50)
 - Respiratory intervention every 1-2h ⁽⁴⁷⁾
 - Short-acting beta-agonist every 1 - 2h or continuous ⁽³³⁾
 - **Episode Day 4, One**:
 - **ACUTE, One**:
 - **Responder**, discharge expected today if clinically stable last 12h, **All**: ⁽³⁹⁾
 - Wheezing, resolved or resolving
 - O₂ sat within acceptable limits ⁽⁴⁰⁾
 - PEF or forced expiratory volume at one second (FEV₁) within acceptable limits or no symptoms of airway obstruction ^(22, 40, 41)
 - Vital signs stable or within acceptable limits ⁽⁴⁰⁾
 - No complication or active comorbidity relevant to this episode of care ⁽⁴²⁾
 - **Non-responder**, not clinically stable for discharge and exceeds episode days at the Acute level of care for this condition, **One**:
 - Use Extended Stay subset
 - Refer for secondary review
 - **INTERMEDIATE, One**:
 - **Non-responder**, not clinically stable for discharge and exceeds episode days at the Intermediate level of care for this condition, **One**:
 - Use Extended Stay subset
 - Refer for secondary review

● **CRITICAL, One:**

- **Non-responder**, not clinically stable for discharge and exceeds episode days at the Critical level of care for this condition, **One:**
 - Use **Extended Stay subset**
 - Refer for secondary review

Discharge, Level of Care, One: ⁽⁴⁹⁾● **HOME, All:**

- Level of care appropriateness, **Both:**
 - Home environment safe and accessible ⁽⁵⁰⁾
 - Patient or caregiver demonstrates ability to manage care
- Asthma discharge planning, **All:** ⁽⁵¹⁾
 - Provide comprehensive written discharge and teaching instructions including asthma action plan ⁽⁵²⁾
 - Delivery device, ≥ **One:**
 - Choose appropriate delivery device
 - Assessment of delivery device technique for all medications
 - Reinforcement of correct technique
 - Review technique with PEF meter
 - Consider referral for case or care management if poorly controlled or frequent visits to the ED
 - Education on the importance of, **All:**
 - Strict medication adherence
 - Schedule follow-up care ^(53, 54)
 - Precipitating factors, ≥ **One:**
 - Identify factors precipitating exacerbation
 - Develop strategies to avoid precipitating factors ⁽⁵⁵⁾
 - Asthma self-management
 - Provide a written asthma action plan or review the patients existing plan
 - Instruction on recognition of signs of recurrence and worsening of asthma
 - Influenza vaccine
 - Medication reconciliation ⁽⁵⁶⁾
 - Follow-up care planned
 - Identify and address transportation needs

● **HOME CARE, All:**

- Level of care appropriateness, **All:**
 - Home environment safe and accessible ⁽⁵⁰⁾
 - Patient or caregiver willing and able to learn care
 - Outpatient management contraindicated or unavailable ^(57, 58)
 - Skilled services, ≥ **One:** ⁽⁵⁹⁾
 - Skilled nursing ⁽⁶⁰⁾
 - Skilled therapy (PT, OT, or ST)
 - Behavioral health ⁽⁶¹⁾
- Asthma discharge planning, **All:** ⁽⁵¹⁾
 - Provide comprehensive written discharge and teaching instructions including asthma action plan ⁽⁵²⁾
 - Delivery device, ≥ **One:**
 - Choose appropriate delivery device
 - Assessment of delivery device technique for all medications
 - Reinforcement of correct technique
 - Review technique with PEF meter
 - Consider referral for case or care management if poorly controlled or frequent visits to the ED
 - Education on the importance of, **All:**
 - Strict medication adherence
 - Schedule follow-up care ^(53, 54)
 - Precipitating factors, ≥ **One:**

- Identify factors precipitating exacerbation
- Develop strategies to avoid precipitating factors ⁽⁵⁵⁾
- Asthma self-management
- Provide a written asthma action plan or review the patients existing plan
- Instruction on recognition of signs of recurrence and worsening of asthma
- Influenza vaccine
- Medication reconciliation ⁽⁵⁶⁾
- Follow-up care planned
- Identify and address transportation needs
- **BEHAVIORAL HEALTH, Both:**
 - Level of care appropriateness, **One:**
 - Support systems available ⁽⁶²⁾
 - Skilled treatment, **≥ One:**
 - Clinical assessment ⁽⁶³⁾
 - Individual, group, or family counseling
 - Psychiatric, substance, or medication evaluation ⁽⁶⁴⁾
- **SKILLED MEDICAL OR THERAPY, Both:**
 - Level of care appropriateness, **All:**
 - Medical practitioner, NP, PA assessment or oversight $\geq 1x/wk$
 - Treatment precluded in a lower level
 - Services, **≥ One:**
 - Medical and skilled nursing interventions or assessment 1-2x/24h ⁽⁶⁰⁾
 - Therapy, **All:**
 - Goal directed therapy with rehabilitation potential ^(65, 66)
 - Functional impairments requiring at least supervision ⁽⁶⁷⁾
 - Able to tolerate 1h/d of skilled therapy $\geq 5d/wk$
 - Complete prior to facility transfer, **All:**
 - Medication reconciliation ⁽⁵⁶⁾
 - Obtain and complete forms for facility
 - Obtain discharge summary and transmit to facility and medical practitioner
 - Arrange transportation
- **SUBACUTE MEDICAL OR THERAPY, Both:**
 - Level of care appropriateness, **All:**
 - Medical practitioner, NP, PA assessment or oversight $\geq 2x/wk$
 - Treatment precluded in a lower level
 - Services, **≥ One:**
 - Medical and skilled nursing interventions or assessment 4h/24h ⁽⁶⁰⁾
 - Therapy, **All:**
 - Goal directed therapy with rehabilitation potential ^(65, 66)
 - ≥ 2 functional impairments requiring at least minimum assistance ⁽⁶⁷⁾
 - Able to tolerate 2-3h/d of skilled therapy $\geq 5d/wk$ and ≥ 1 therapy discipline
 - Complete prior to facility transfer, **All:**
 - Medication reconciliation ⁽⁵⁶⁾
 - Obtain and complete forms for facility
 - Obtain discharge summary and transmit to facility and medical practitioner
 - Arrange transportation
- **SUBACUTE REHAB, Both:**
 - Level of care appropriateness, **All:**
 - Medical practitioner, NP, PA assessment or oversight $\geq 3x/wk$
 - Treatment precluded in a lower level
 - Services, **All:**
 - Goal directed therapy with rehabilitation potential ^(65, 66)
 - ≥ 2 functional impairments requiring at least minimum assistance ⁽⁶⁷⁾

- Able to tolerate 2-3h/d of skilled therapy ≥ 5d/wk and ≥ 2 therapy disciplines
- Complete prior to facility transfer, **All:**
 - Medication reconciliation ⁽⁵⁶⁾
 - Obtain and complete forms for facility
 - Obtain discharge summary and transmit to facility and medical practitioner
 - Arrange transportation
- **ACUTE REHAB, Both:**
 - Level of care appropriateness, **Both:**
 - Medical practitioner, NP, PA assessment or oversight ≥ 3x/wk
 - Services, **All:**
 - Goal directed therapy with rehabilitation potential ^(65, 66)
 - ≥ 2 functional impairments requiring at least minimum assistance ⁽⁶⁷⁾
 - Able to tolerate ≥ 15h/wk of skilled therapy and ≥ 2 therapy disciplines ⁽⁶⁸⁾
- Complete prior to facility transfer, **All:**
 - Medication reconciliation ⁽⁵⁶⁾
 - Obtain and complete forms for facility
 - Obtain discharge summary and transmit to facility and medical practitioner
 - Arrange transportation
- **LONG TERM ACUTE CARE, Both:**
 - Level of care appropriateness, **Both:**
 - Treatment precluded in a lower level
 - In lieu of acute or continued hospitalization or failed lower level of care, ≥ **One:**
 - Primary condition or illness and treatment of active comorbid condition(s) ⁽⁶⁹⁾
 - Ventilator dependent ≥ 6h/d and weaning planned ^(70, 71)
 - Acute and comorbid conditions requiring prolonged hospitalization ⁽⁷²⁾
- Complete prior to facility transfer, **All:**
 - Medication reconciliation ⁽⁵⁶⁾
 - Obtain and complete forms for facility
 - Obtain discharge summary and transmit to facility and medical practitioner
 - Arrange transportation

Utilization Benchmarks

Condition or Procedure	% Pd Obs	LOS (days)	Type
202 BRONCHITIS AND ASTHMA WITH CC/MCC	n/a	2.9	CMS GMLOS
203 BRONCHITIS AND ASTHMA WITHOUT CC/MCC	n/a	2.2	CMS GMLOS
ASTHMA	43%	2.9	InterQual

Percent Paid as Observation indicates the final payment status as a percentage for patients hospitalized with the condition. It may not correspond to the status assigned at admission. Note that patients initially placed in Observation status who are converted to inpatient status (common) are included in the inpatient category, and that patients initially admitted to inpatient status who are converted to observation status (uncommon) are included in the Observation category.

Notes:**1:**

The CMS Two-Midnight Rule generally requires that the medical record support a patient's need for hospital-level services **AND** that the patient has a hospital stay of at least two midnights. Short-stay exceptions to this rule are defined below. Application of InterQual® criteria can help an organization comply with the medical necessity requirements of the two-midnight rule.

For those who are impacted by the CMS Two-Midnight Rule, follow the steps below:

1. **Determine the need for hospital-based services by applying criteria.** If a patient meets criteria at any level of care, they have met the medical necessity requirement of the rule.
2. **Determine whether the patient is an inpatient or assign observation status** depending on which level of care is met and the expected length of stay. Based on the presentation of the patient, evidence-based criteria such as InterQual® can support the provider's expectation for length of stay.
 - **When care is expected to span two midnights** and the patient has met criteria at the Acute, Intermediate or Critical level of care, an inpatient status may be appropriate.

NOTE: In the event of an "Unforeseen Circumstance" per CMS resulting in a shorter stay than initially expected, a status of inpatient may still be appropriate:

- Leaving against medical advice (AMA)
- Unexpected death
- Transfer of the patient
- Unexpected clinical improvement
- Election of hospice care

- **When care is expected to span less than two midnights** but Acute, Intermediate, or Critical criteria are met, it may be appropriate to initially assign a status of observation.

NOTE: CMS has defined the following exceptions as being appropriate for inpatient status regardless of length of stay:

- Mechanical Ventilation, established during current hospitalization
- Surgical procedure or intervention found on the CMS Inpatient Only List

- **When it is undecided as to whether care will span two midnights**, the reviewer may use the level of care met as a guide to determine whether the patient is appropriate as an inpatient or observation status. The criteria consider multiple factors, including clinical severity, comorbidity, and risk, when differentiating between the Observation and Acute levels of care. These factors impact the length of stay. A patient meeting criteria at the Acute levels of care can generally be expected to have a length of stay of greater than two midnights and thus could be assigned to inpatient status. Patients who meet criteria at the Observation level of care are more appropriate for observation status.

(Centers for Medicare & Medicaid Services, Title 42; Part 412; Subpart A - Admissions. 2023; Centers for Medicare & Medicaid Services. Title 42; Part 422; Subpart C-Benefits and Beneficiary Protections. 2023; Centers for Medicare & Medicaid Services, In: BFCC-QIO 2 Midnight Claim Review Guideline. 2022)

2:

Initial Review criteria are a level of care determination tool, intended to be used as **real-time** decision support in the emergency department to identify if observation or inpatient hospital level services are warranted. They help the reviewer determine whether a patient is appropriate for observation or inpatient admission at the time a decision to admit the patient is being made. Initial Review criteria evaluate **only data that are available at the time the decision is being made**. This may include previously provided interventions or the results of laboratory, imaging, and other tests. Initial Review may be appropriate when bridge or holding orders (e.g., admit to telemetry) are in place. These orders are intended to address the patient's needs until full treatment and medication orders are written.

Initial Review criteria **should not be used** retrospectively or once the decision to admit has been made (e.g., when full treatment and medication orders have been written). If the reviewer has sufficient information to complete an Episode Day 1 review, an Initial Review need not be completed.

Note: While Initial Review enables identification of the level of care, it is not intended to, and should not be used as a substitute for an Episode Day 1 review, which will generally include specific, evidence-based interventions and

intensity of service requirements.

For further clarification, see the Acute Level of Care Review Process.

3:

Expected progress after 12-24h

- Peak expiratory flow (PEF) or forced expiratory volume in one second (FEV₁) returning to baseline
- Vital signs normalizing (heart rate, respiratory rate), no longer requires accessory muscle breathing
- Can speak in complete sentences without dyspnea
- Weaning off supplemental O₂
- Decreased wheezing/dyspnea
- O₂ saturation improving
- Afebrile

Consider:

- Decreasing frequency of nebulized treatments or metered dose inhaler (MDI)
- Decreasing dose or frequency of corticosteroids
- Discharge planning

4:

Care Facilitation

Potential risk for readmission

- Knowledge deficit or non-adherence with condition management identified (**Home Care, Readmission reduction program**)

Social Determinants of Health (SDOH)

Social concerns may impact the decision to admit, care coordination during hospital stay, as well as discharge destination. The following social determinants of health should be evaluated on admission to determine which factors are relevant to an individual patient and condition. This list may not be all inclusive and all relevant SDOH factors should be considered. Any SDOH factors determined to be relevant to a given patient should be planned for throughout the hospital stay.

Alcohol or substance use

- Care coordination with outpatient medical or behavioral health providers
- Information should be given about community resources and programs
- Referral to addiction social support systems

Domestic violence and sexually abused individuals

- Community resources and legal information should be given
- Create a long-term safety plan or make appropriate referral

End of life planning, arrange palliative care or hospice

- Consider palliative care for patients with end stage disease
- Consider hospice care for terminally ill patients with a life expectancy of less than 6 months

Financial resources strained or limited

- For patients with income insecurity, identify income-support resources (Pottie et al., CMAJ 2020, 192: E240-E54)
- Consider direct income assistance benefits and indirect income assistance programs (i.e., programs that improve access to basic living necessities such as provision of food, daycare, and fuel or rent supplements)
- Consider referral to Area Agencies on Aging (AAA) for information on nutrition and meals programs, homemaking services, transportation resources
- For food insecurity, provide information about local food banks and Supplemental Nutrition Assistance Program (SNAP) incentive programs (Berman et al., Pediatr Rev 2018, 39: 235-46)

Functional or ADL impairment

- Consider an alternate level of care
- If patient is unable to make decision for self, refuses treatment, or demands to be discharged without a rational reason, a capacity evaluation may be needed to ensure patient's ability to make decisions for self

- Information should be given to patient and family about Advance Directive or Power of Attorney

Mental health co-morbidities involved

- Appointment with primary care provider or behavioral health specialist upon discharge
- Care coordination with outpatient behavioral health providers
- Community resources should be given

Risk for readmission

- Consider home care services
- Consider transfer to an alternative level of care or readmission reduction program
- Provide patient and family with information on Disease Management programs
- Consider telehealth and mobile health interventions

Risk for treatment nonadherence

- For patients with poor insight into illness or noncompliance with treatment, introduce peer support groups to help improve service user experience and quality of life (National Institute for Health and Care Excellence (NICE). Psychosis and schizophrenia in adults: prevention and management. CG178. London: NICE; 2014.)
- In-home intervention to prevent medical relapse, or readmission, and to enhance medication compliance
- For patients at risk of medication nonadherence, consider patient education and behavioral support initiatives such as text message reminders to take a medication and electronic medication dispensers.

Environmental exposure to toxic substances and other physical hazards

- For tobacco use and/or exposure information should be given about tobacco use cessation and the risk of second-hand smoke
- For lead poisoning, provide community resources for primary prevention and secondary prevention strategies (i.e., blood lead level testing and follow up) (Centers for Disease Control and Prevention, Lead Poisoning Prevention. 2019)

Unstable living environment or homelessness expected upon discharge

- Health insurance application
- Homeless shelter application and information should be given to patient
- Referral to walk-in state or federal sponsored outpatient programs
- Ensure access to local housing coordinator or case manager for immediate link to permanent supportive housing and coordinated access system (Pottie et al., CMAJ 2020, 192: E240-E54)
- For individuals with homelessness and the greatest service needs (i.e., serious mental illness, history of hospitalizations, functional impairment) consider Intensive Case Management Intervention, Assertive Community Treatment, or other Intensive Community-Based Treatments (Ponka et al., PLoS One 2020, 15: e0230896)

Unsafe home environment

- If home environment is unsafe, consider a skilled nursing facility (SNF)
- For patients at risk for falls, provide information on community-based fall prevention programs through the National Council on Aging

5:

Hemodynamically stable patients who require at least 6 hours, and for certain conditions, up to 48 hours, of treatment or assessment pending a decision regarding the need for additional care. Observation level of care services may be provided in designated observation units or on a hospital floor.

Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions consider a variety of factors, such as clinical judgment, medical necessity, payer policies and contracts, and state and national regulations.

6:

These risk factors have been shown to increase the risk of asthma-related mortality and asthma severity (Global Initiative for Asthma (GINA), Global Strategy for Asthma Management and Prevention. 2023; Pena et al., Emerg Med Clin North Am 2017, 35: 625-45; National Heart, Lung, and Blood Institute. Expert Panel Report 3: Guidelines for the

Diagnosis and Management of Asthma - Full Report 2007). Patients who respond to initial treatment and have any of these risk factors may require continued observation for risk of decompensation, to address adherence issues, and to ensure a support system is in place prior to discharge.

7:

Comorbidities include:

- Cardiovascular (including arrhythmias)
- Chronic lung disease (e.g., COPD)
- Chronic psychiatric (e.g., anxiety, major depressive disorder)
- Gastroesophageal reflux disease (GERD)
- Diabetes
- Dysfunctional breathing
- Obesity
- Obstructive sleep apnea

8:

Patients with asthma who have a chronic comorbid disease have worse medical outcomes than those without and may be at increased risk for asthma-related death (Global Initiative for Asthma (GINA), Global Strategy for Asthma Management and Prevention. 2023; Mahdavian et al., Can Respir J 2018, 2018: 6460379). The following comorbid conditions in patients with asthma have been associated with increased hospitalization, length of stay, and/or ICU admissions (Kaplan et al., NPJ Prim Care Respir Med 2020, 30: 36; Tay et al., Respirology 2016, 21: 1384-90):

- Cardiovascular (including arrhythmias)
- Chronic lung disease (e.g., COPD)
- Chronic psychiatric (e.g., anxiety, major depressive disorder)
- Diabetes
- Dysfunctional breathing
- Gastroesophageal reflux disease (GERD)
- Obesity
- Obstructive sleep apnea

These conditions may complicate asthma diagnosis as they can have overlapping symptoms, which can lead to misdiagnosis and mistreatment. Comorbid conditions may also lead to polypharmacy which can have a negative impact on asthma adherence and control. Treatment of asthma and comorbidities may have a bidirectional effect where treatment for one has a negative effect on the other; for example, high dose inhaled corticosteroids impacts blood glucose in patients with diabetes. (Kaplan et al., NPJ Prim Care Respir Med 2020, 30: 36).

9:

Patients who have had more than 1 severe asthma exacerbation in the past year or have ever been intubated due to asthma are more at risk for poor asthma outcomes and asthma related deaths (Global Initiative for Asthma (GINA), Global Strategy for Asthma Management and Prevention. 2023).

10:

Current oral steroid use is a marker of asthma severity and may indicate an increased risk for asthma-related mortality (Global Initiative for Asthma (GINA), Global Strategy for Asthma Management and Prevention. 2023). These patients may require continued monitoring at the Observation level of care. Although the literature does not define a specific time for "recently stopped" oral steroid use in patients greater than 5 years of age, the definition of "within the past month" is based upon best practice as determined by InterQual's® external peer reviewers. This determination is based on an iterative process involving multiple clinicians with diverse expertise in varied practices and geographic settings.

11:

Patients who have a poor adherence to maintenance treatment medication, such as inhaled corticosteroids, and patients who do not have or do not adhere to their asthma action plan are at higher risk for asthma exacerbation and/or poor asthma outcomes (Global Initiative for Asthma (GINA), Global Strategy for Asthma Management and

Prevention. 2023).

12:

Instruction: Treatment is inclusive of interventions provided in a medical practitioner's office, outpatient clinic (e.g., urgent care center), or emergency department.

13:

Initial treatment for an acute asthma exacerbation consists of inhaled short-acting beta-agonists and systemic corticosteroids. Epinephrine (for anaphylaxis), inhaled ipratropium bromide, and/or intravenous magnesium may also be used in some cases (Global Initiative for Asthma (GINA), Global Strategy for Asthma Management and Prevention. 2023; National Heart, Lung, and Blood Institute. Expert Panel Report 3: Guidelines for the Diagnosis and Management of Asthma - Full Report 2007).

14:

In patients who present with acute asthma exacerbation, frequent reassessments after the initial management with short-acting beta-agonist (SABA) and titration of further treatments based on the clinical response are recommended. If peak expiratory flow or forced expiratory volume in 1 second is being used to evaluate the patient, the recommendation is to repeat the measurement 1 hour after the initial treatment to assess the response to bronchodilator treatments (Global Initiative for Asthma (GINA), Global Strategy for Asthma Management and Prevention. 2023).

15:

Treatment with both a short-acting beta-agonist (SABA) and ipratropium bromide were associated with fewer hospitalizations and greater improvement in peak expiratory flow and forced expiratory volume than a SABA alone in adults and children with a moderate to severe asthma exacerbation (Global Initiative for Asthma (GINA), Global Strategy for Asthma Management and Prevention. 2023; Kirkland et al., Cochrane Database Syst Rev 2017, 1: CD001284).

16:

Initial treatment for an acute asthma exacerbation includes:

- Short-acting beta-agonists ≥ 2 doses
- Ipratropium bromide (unless contraindicated) ≥ 2 doses
- Systemic corticosteroids (including PO)

17:

Persistent refers to wheezing after initial treatment.

18:

Baseline refers to either the patient's normal baseline or a newly established baseline. In the absence of documentation, a patient's baseline status may be presumed to be normal.

19:

Mild to moderate asthma exacerbation may have oxygen saturation between 90-95%(0.90-0.95). Lower oxygen level may be seen in severe exacerbation (Global Initiative for Asthma (GINA), Global Strategy for Asthma Management and Prevention. 2023)

20:

Forced expiratory volume (FEV₁) is the maximum amount of air that can be forcefully exhaled in 1 second. It is then converted to a percentage predicted. FEV₁ is a marker for the degree of obstruction caused by asthma. Although peak expiratory flow (PEF) may not be as reliable as FEV₁, PEF is another way to measure degree of obstruction and can be helpful in evaluating severity of asthma and response to asthma treatments. With training, patients can perform and monitor PEF with a peak flow meter at home, and PEF is often included as a part of asthma action plan. There are predicted FEV₁ and PEF values available based on height, weight, sex, and/or race that can be used

as reference when evaluating the severity of asthma exacerbation. Patient's personal best values, if available, can also be used (Global Initiative for Asthma (GINA), Global Strategy for Asthma Management and Prevention. 2023; National Heart, Lung, and Blood Institute. Expert Panel Report 3: Guidelines for the Diagnosis and Management of Asthma - Full Report 2007). FEV₁ and PEF measurements corresponds to the severity of asthma exacerbation as follows (National Heart, Lung, and Blood Institute. Expert Panel Report 3: Guidelines for the Diagnosis and Management of Asthma - Full Report 2007):

- $\geq 70\%$ (0.70) predicted/personal best = Mild exacerbation
- 40% to 69%(0.40 to 0.69) predicted/personal best = Moderate exacerbation
- $< 40\%$ (0.40) predicted/personal best = Severe exacerbation

21:

Guidelines strongly recommend the measurement of lung function. These measurements can be more reliable indicators of severity than symptoms. If possible, peak expiratory flow (PEF) or forced expiratory volume in 1 second (FEV₁) should be recorded prior to the initiation of treatment and then at one hour after treatment and at intervals until a response to treatment is noted (Global Initiative for Asthma (GINA), Global Strategy for Asthma Management and Prevention. 2023; National Heart, Lung, and Blood Institute. Expert Panel Report 3: Guidelines for the Diagnosis and Management of Asthma - Full Report 2007).

22:

- Persistent dyspnea
- Prefers sitting to lying
- Talks in phrases - unable to complete sentences

23:

In an effort to address inequities in medicine and improve appropriate access to care, multiple medical societies have been reevaluating the use of adjustments made for a patient's race and updating their calculators and reference equations when reporting diagnostic results.

In 2023, the American Thoracic Society issued updated guidance on reporting of pulmonary function tests (PFTs), advocating for a race-neutral average reference equation to determine the baseline (i.e., predicted normal) for a given patient (Bhakta et al., Am J Respir Crit Care Med 2023, 207: 978-995). The reference equation previously included lower predicted normal values for patients of certain races and ethnicities (in the US, for Black and Asian patients); a practice which has not been found to have biological validity and may lead to misclassification of diagnoses and disease severity in racial or ethnic minorities (Vyas et al., N Engl J Med 2020, 383: 874-882). Since not all spirometer manufacturers and PFT laboratories have switched over to the new race-neutral average reference equation, clinical judgement and consideration of patient-specific factors should be used when interpreting PFT results and deciding whether patients with borderline reported PFT values in the presence of concerning symptoms or signs should have further evaluation or treatment.

24:

Vital signs are considered sustained findings when they are abnormal for 2 or more readings greater than 15 minutes apart or are lasting at least 15 minutes. This excludes an isolated reading, a transient abnormal measurement, or a finding that requires urgent treatment (e.g., severe hypoxemia, ventricular arrhythmia, hypotension). Application of criteria for a vital sign finding should be based on confirmatory measurements repeated at regular intervals. Events that are infrequent or vital sign abnormalities that have resolved with outpatient treatment are not considered to be sustained (e.g., resolution of hypertension following anti-hypertensive therapy, resolution of tachycardia following rehydration).

Note: The definition of sustained reflects the opinion of InterQual's® external peer reviewers. This criteria point is based upon current best practice and is the product of an iterative process involving multiple clinicians with diverse expertise in varied geographic settings.

25:

Respiratory rate can be used to assess the severity of asthma exacerbation. Increased respiratory rate may be seen in any asthma exacerbation, but a respiratory rate of greater than 30 breaths/min describes severe exacerbation (Global Initiative for Asthma (GINA), Global Strategy for Asthma Management and Prevention. 2023; National Heart, Lung, and Blood Institute. Expert Panel Report 3: Guidelines for the Diagnosis and Management of Asthma - Full

Report 2007). Higher levels of care may be necessary for patients with a persistent respiratory rate of greater than 30 breaths/min.

26:

Hemodynamically unstable patients (or those with the potential to become unstable) who require treatment, assessment, or intervention every 1 to 2 hours.

Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions consider a variety of factors, such as clinical judgment, medical necessity, payer policies and contracts, and state and national regulations.

27:

Signs of worsening respiratory failure include the inability to speak, altered mental status, intercostal retractions, increasing fatigue, and rising PCO_2 . In life-threatening exacerbations, additional treatments may be given in attempt to avoid intubation including noninvasive positive pressure ventilation (NIPPV) and heliox. Avoiding intubation is a reasonable goal, but once it is deemed necessary, it should not be delayed (Global Initiative for Asthma (GINA), Global Strategy for Asthma Management and Prevention. 2023; National Heart, Lung, and Blood Institute. Expert Panel Report 3: Guidelines for the Diagnosis and Management of Asthma - Full Report 2007).

28:

Status asthmaticus, marked by continuous bronchospasm refractory to standard treatment, is a severe, life-threatening form of asthma. Inadequate gas exchange results in hypoxia and hypercapnia. Additional findings may include agitation, arrhythmias, cyanosis, hypotension, and exhaustion (Scottish Intercollegiate Guidelines Network (SIGN), British guideline on the management of asthma. 2019).

29:

Asthma usually results in an increased respiratory drive which often leads to an arterial PCO_2 that is below normal. Normal or increased PCO_2 in a patient with asthma exacerbation could indicate impending respiratory failure. Arterial blood gas is not always necessary but may be considered in patients who are not responding to initial treatment or are worsening despite treatment. In patients with chronic CO_2 retention hospitalized for acute respiratory illness, their PCO_2 values can be compared with baseline values, when available, and other signs and symptoms of respiratory distress can be used to determine the appropriate level of care (e.g., worsening dyspnea, high respiratory rate, decreased oxygen saturation, confusion, drowsiness) (Global Initiative for Asthma (GINA), Global Strategy for Asthma Management and Prevention. 2023; McKeever et al., Thorax 2016, 71: 210-5).

30:

Use of accessory muscles for breathing is an indicator of severe asthma (Global Initiative for Asthma (GINA), Global Strategy for Asthma Management and Prevention. 2023). Accessory muscles used for breathing include the neck (e.g., sternocleidomastoid, suprasternal), intercostals, and thoracic.

31:

The National Heart, Lung, and Blood Institute guidelines on asthma recommend admitting patients to the intensive care unit when post-treatment peak expiratory flow (PEF) or forced expiratory volume at 1 second (FEV_1) is less than 40%(0.40). Global Initiative for Asthma (GINA) does not specify the level of care for this group of patients but does recommend that hospitalization is necessary at this post-treatment PEF level (Global Initiative for Asthma (GINA), Global Strategy for Asthma Management and Prevention. 2023; National Heart, Lung, and Blood Institute. Expert Panel Report 3: Guidelines for the Diagnosis and Management of Asthma - Full Report 2007).

32:

Systemic corticosteroid is recommended to be given promptly for most patients presenting to the hospital with asthma exacerbation. Oral steroids can be given at the onset and continued for 5-7 days. The oral route may be preferred because it is quicker and less invasive. Intravenous dosing may be needed in patients with severe asthma symptoms who may not be able to take oral medications safely (Global Initiative for Asthma (GINA), Global Strategy for Asthma Management and Prevention. 2023).

33:

Instruction: This criterion applies to the administration of a bronchodilator via nebulizer or metered-dose inhalers (MDI) with or without a spacer. MDIs produce outcomes that are at least equivalent to nebulizers. The use of a spacer may improve medication delivery via an MDI for patients who have difficulty with proper technique. Spacers may have some advantages compared to nebulizers for children with acute asthma. The use of an inhaler with a spacer is also recommended when viral transmission via nebulizer can lead to increased risk of COVID-19 (Global Initiative for Asthma (GINA), Global Strategy for Asthma Management and Prevention. 2023; National Heart, Lung, and Blood Institute. Expert Panel Report 3: Guidelines for the Diagnosis and Management of Asthma - Full Report 2007; Cates et al., Cochrane Database Syst Rev 2013: CD000052).

34:

Heliox, a combination of helium and oxygen administered through an oxygen delivery system, has been used in the treatment of severe asthma. Helium improves the flow of oxygen through restricted airways due to its lower density. Its use is not routinely recommended but may be considered for patients with refractory symptoms (Global Initiative for Asthma (GINA), Global Strategy for Asthma Management and Prevention. 2023; Craig et al., Cochrane Database Syst Rev 2020, 8: CD012977).

35:

Noninvasive positive pressure ventilation (NIPPV), also known as NIV, provides respiratory support by application of a tightly fitting facial mask, nasal mask, or helmet rather than an endotracheal tube or tracheostomy. In some cases, the use of NIPPV can avoid the need for endotracheal intubation and decreases the risk of barotrauma, lung injury, and/or infection. NIPPV is commonly delivered by a bilevel positive airway pressure ventilator (BiPAP), a continuous positive airway pressure device (CPAP), or a mechanical ventilator. Supplemental oxygen can be delivered at concentrations approximating 100%(1.0).

The decision to provide respiratory support via NIPPV, and the modality to provide NIPPV, is based upon the patient's specific clinical findings (e.g., medical condition leading to respiratory failure, underlying comorbidities, clinical progression, improvement). Examples of NIPPV devices are volumetric (i.e., deliver a defined volume), barometric (i.e., deliver a defined pressure), and combined (i.e., deliver defined volumes and pressure). The terminology for NIPPV delivery systems may differ between equipment manufacturers and provider organizations. InterQual® criteria do not differentiate between the different NIPPV modalities.

Whether or not high-flow nasal cannula (HFNC) should be classified as a component of NIPPV is unclear, and there is conflicting evidence in the medical literature. Where HFNC is an appropriate modality, InterQual® defines this in a specific criteria point. As such, NIPPV does not include HFNC (Hackett, A., PulmCCM 2018; Nardi et al., F1000Res 2017, 6: 290; Osadnik et al., Cochrane Database Syst Rev 2017, 7: CD004104; Allison and Winters, Emerg Med Clin North Am 2016, 34: 51-62; Gregoretti et al., Crit Care Clin 2015, 31: 435-57).

36:

There is limited evidence on the use of noninvasive positive pressure ventilation (NIPPV) in the treatment of severe asthma, but it may be attempted with close monitoring in some patients (Global Initiative for Asthma (GINA), Global Strategy for Asthma Management and Prevention. 2023; Paton, British guideline on the management of asthma. SIGN 158. 2019).

37:

The CMS Two-Midnight Rule generally requires that the medical record support a patient's need for hospital level services **AND** that the patient has a hospital stay of at least two midnights.

For patients who are in observation status and do not meet Observation responder criteria on Episode Day 2, apply Episode Day 2 criteria at the Observation, Acute, Intermediate or Critical level of care in the appropriate subset to demonstrate the continued need for hospital-based services.

NOTE: For review of a surgical patient, Post-Operative Day 1 equates to Episode Day 2.

(Centers for Medicare & Medicaid Services, Title 42; Part 412; Subpart A - Admissions. 2023; Centers for Medicare & Medicaid Services. Title 42; Part 422; Subpart C-Benefits and Beneficiary Protections. 2023; Centers for Medicare & Medicaid Services, In: BFCC-QIO 2 Midnight Claim Review Guideline. 2022)

38:

Expected progress after 48h

- Activities at or close to baseline
- Wheezing or dyspnea resolved or resolving
- O₂ saturation at or close to baseline
- Vital signs stable or within acceptable limits
- Preparing for discharge within the next 24 hours

If no, consider (Global Initiative for Asthma (GINA), Global Strategy for Asthma Management and Prevention. 2023):

- IV magnesium for patients with severe exacerbation
- Heliox for patients that are not responding to standard therapy
- Imaging or diagnostic testing to rule out pneumonia, pulmonary embolism, heart failure, arrhythmia, pneumothorax, or pleural effusion
- Blood gas if the patient is deteriorating

39:

Selection of this criteria point indicates that the patient is responding to treatment and is clinically stable for transfer or discharge. To determine the most appropriate post-acute level of care, see discharge criteria.

40:

When a criteria point states "within acceptable limits," it refers to either the patient's normal baseline, a newly established baseline, or parameters that the medical practitioner determines are acceptable.

41:

Recommendations differ on the acceptable threshold for peak expiratory flow (PEF) or forced expiratory volume at 1 second (FEV₁) values for safe discharge. Guidelines from the National Heart, Lung, and Blood Institute state that post-treatment PEF or FEV₁ of greater than or equal to 70%(0.70) is generally an appropriate cutoff to discharge patients from the ED or hospital. Patients with post-treatment PEF or FEV₁ less than that are recommended to be assessed individually for possible discharge based on risk factors. Global Initiative for Asthma (GINA) classifies post-treatment PEF or FEV₁ of greater than 60-80%(0.60-0.80) as a moderate exacerbation and recommends considering discharging patients from the emergency department (ED) if post-treatment PEF or FEV₁ is greater than or equal to 60%(0.60) after considering risk factors and follow-up availability (Global Initiative for Asthma (GINA), Global Strategy for Asthma Management and Prevention. 2023; National Heart, Lung, and Blood Institute. Expert Panel Report 3: Guidelines for the Diagnosis and Management of Asthma - Full Report 2007).

42:

Relevant complications or comorbidities are conditions that require active management during an inpatient stay.

43:

Hemodynamically stable patients who require treatment, assessment, or intervention every 4 to 8 hours.

Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions consider a variety of factors, such as clinical judgment, medical necessity, payer policies and contracts, and state and national regulations.

44:

Hemodynamically stable patients who require treatment, assessment, or intervention every 2 to 4 hours.

Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions consider a variety of factors such as clinical judgment, medical necessity, payer policies and contracts, and state and national regulations.

45:

Oxygen therapy is the administration of oxygen at concentrations greater than ambient air (room air: 21%(0.21)) with the intent of treating and/or preventing the symptoms and manifestations of hypoxia. The oxygen

concentration or percentage (FiO₂) delivered varies with the manufacturer's design, oxygen flow rate, the patient's respiratory rate, and tidal volume. Actual inspired FiO₂ with nasal cannula varies significantly with the patient's minute ventilation and pattern of breathing. For patients at risk for CO₂ retention, where a precise inspired FiO₂ is required, the Venturi mask is the preferred method. In general, patients who are very ill or have respiratory disease may require considerably higher flow rates to achieve the desired FiO₂. The following are estimates of O₂ delivered at the associated flow rates:

LOW-FLOW SYSTEMS

Nasal cannula

Room air 21%(0.21)

1L 24%(0.24) 4L 36%(0.36)

2L 28%(0.28) 5L 40%(0.40)

3L 32%(0.32) 6L 44%(0.44)

Simple oxygen masks

- 35-50%(0.35-0.50) FiO₂ (5-10 L/min)
- Flow rates are usually maintained at 5 L/min or more to avoid accumulation of CO₂ in the mask

Venturi masks

- 24-50%(0.24-0.50) FiO₂ (4-12 L/min)

Partial rebreathing masks

- 40-70%(0.40-0.70) FiO₂ (6-10 L/min)

Non-rebreathing masks

- 60-80%(0.60-0.80) FiO₂ (minimum flow of 10 L/min)

HIGH-FLOW SYSTEMS

Air-entrainment masks/nebulizers

- 24-40%(0.24-0.40) FiO₂

Heated, humidified high-flow concentrating nasal cannula (HFNC)

- 24-100%(0.24-1.0) FiO₂

46:

Guidelines recommend supplemental oxygen use in acute asthma exacerbation. Target oxygen saturation levels differ between guidelines, but the latest guideline from Global Initiative for Asthma (GINA) in 2023 recommended keeping oxygen saturation at 93%(0.93) to 95%(0.95) (Global Initiative for Asthma (GINA), Global Strategy for Asthma Management and Prevention. 2023; Paton, British guideline on the management of asthma. SIGN 158. 2019; National Heart, Lung, and Blood Institute. Expert Panel Report 3: Guidelines for the Diagnosis and Management of Asthma - Full Report 2007).

47:

Respiratory interventions include ventilator/noninvasive positive pressure ventilation (NIPPV) setting changes, chest physiotherapy, suctioning, and nebulizer or metered-dose inhaler (MDI) treatments.

48:

The CMS Two-Midnight Rule generally requires that the medical record support a patient's need for hospital level services **AND** that the patient has a hospital stay of at least two midnights.

For patients who are in observation status beyond the second midnight (Episode Day 3) and who do not meet Observation responder criteria, apply Episode Day 3 Inpatient criteria in the appropriate condition-specific or general subset.

Note: Patients who meet criteria **at any of** the Inpatient levels of care **based on medically necessary hospital-based services and supported by clinical documentation would be appropriate for inpatient status**; those who do not meet criteria would be appropriate for secondary review.

(Centers for Medicare & Medicaid Services, Title 42; Part 412; Subpart A - Admissions. 2023; Centers for Medicare & Medicaid Services. Title 42; Part 422; Subpart C-Benefits and Beneficiary Protections. 2023; Centers for Medicare & Medicaid Services, In: BFCC-QIO 2 Midnight Claim Review Guideline. 2022)

49:

The discharge screen is a resource tool and not criteria. Referring to the discharge screen at the initiation of discharge planning is recommended.

50:

The home environment and safety assessment will normally include an evaluation of the patient's pre- and post-hospitalization functional level, physical layout of the home (e.g., entrances and exits, stairs, access to the community), identification of unsafe conditions (e.g., scatter rugs, missing handrails, oxygen use and smoking, lack of fire safety devices, inadequate lighting, heating, and cooling), factors that may trigger symptoms (e.g., secondhand smoke, poor food choices, dysfunctional family dynamics), sanitation hazards (e.g., lack of electricity, running water, refrigeration, inadequate toilet facilities, presence of insects or rodents), and the need for adaptive equipment (e.g., walker, handrails for the tub or shower, elevated toilet seat).

51:

Evidenced-based guidelines recommend that patients utilize a daily diary and perform a periodic self-assessment (can be completed in the practitioner's office) to promote self-management skills and to identify their personal perception of asthma control and perceived satisfaction with care. Patients should follow either a peak expiratory flow (PEF) monitoring-based action plan and/or a symptom monitoring-based plan. Both have been found to be equally beneficial when taught and followed correctly. Furthermore, the self-monitoring plan must be individualized and tailored to meet patient and/or family preferences (Global Initiative for Asthma (GINA), Global Strategy for Asthma Management and Prevention. 2023; National Heart, Lung, and Blood Institute. Expert Panel Report 3: Guidelines for the Diagnosis and Management of Asthma - Full Report 2007).

52:

Ensuring the patient and/or caregiver understands all aspects of the condition and can assume responsibility for self-care is crucial in preventing readmission. Assessing a patient and/or caregiver's level of understanding after education has been provided can be difficult. The teach-back method is an effective way of providing education at the appropriate level and can also be used to assess the learner's comprehension. To use the teach-back method, discuss key points in common terms and avoid using medical jargon or unfamiliar terms. After the education has been delivered, ask the learner to repeat what was learned. Gaps or misinterpretations in the learner's explanation will pinpoint areas where communication may have failed and provide opportunity for clarification.

53:

Patients should be referred to an asthma specialist in the following situations (Global Initiative for Asthma (GINA), Global Strategy for Asthma Management and Prevention. 2023; National Heart, Lung, and Blood Institute. Expert Panel Report 3: Guidelines for the Diagnosis and Management of Asthma - Full Report 2007):

- When there has been a life-threatening asthma exacerbation
- If the goals of asthma therapy are not being met after 3 to 6 months of treatment
- When there are atypical signs and symptoms or difficulties confirming diagnosis of asthma
- If other conditions complicate asthma
- When additional education and guidance is needed (e.g., complications of therapy, problems with adherence, allergen avoidance)
- For consideration of immunotherapy
- If the patient has persistent asthma despite 2 or more medications
- If more than 2 courses of oral systemic corticosteroids are required in 1 year
- If there is a history of an occupational or environmental inhalant or ingested substance contributing to asthma

54:

The first follow up appointment is recommended within the first 2 to 7 days of hospital or emergency department discharge. An appointment with an asthma specialist should also be scheduled when indicated (Global Initiative for Asthma (GINA), Global Strategy for Asthma Management and Prevention. 2023).

55:

There are many triggers or precipitating events leading up to an asthma exacerbation. It is important to control or

avoid those triggers whenever possible. Some suggestions to assist patients in this include (National Heart, Lung, and Blood Institute. Expert Panel Report 3: Guidelines for the Diagnosis and Management of Asthma - Full Report 2007):

- Smoking cessation
- Not smoking in the home or around children
- Encasing your mattress and pillows with dust-mite proof covers
- Decreasing indoor humidity below 60%(0.60)
- Keeping pet fur or hair out of the home
- Keeping all food out of the bedroom
- Keeping food and garbage in closed containers
- Using a vacuum with a high-efficiency particulate air (HEPA) filter or a double-layered bag
- Fixing leaky faucets, pipes, and other water sources
- Cleaning moldy surfaces
- Keeping windows closed during allergy season
- Asking your doctor when to increase medications
- Avoiding wood burning stoves
- Asking your doctor about taking medications before exercise to prevent symptoms
- Avoiding sulfites in food
- Covering your nose and mouth on cold windy days

56:

Medication reconciliation is a formal process or technique used by health care providers and pharmacists to identify the most complete and accurate list of all medications a patient is taking at times of transitions in care (e.g., upon hospital admission, transfer from one unit to another during hospitalization, or discharge from the hospital to home or another facility). The goals of this process are to ensure medication and dosages are appropriate for the patient, resolve discrepancies in drug regimens, and ultimately prevent medication errors and reduce adverse drug events. Medication reconciliation is a Joint Commission National Patient Safety goal. Coordinating information when a patient is transferred to another setting, service, practitioner, or level of care ensures accurate medications are listed. The process is comprised of the following (Hospital: National Patient Safety Goals. 2020):

- Obtain an external list of medications (e.g., medications taken prior to admission)
- Develop a list of current medications and add them to the medical record
- Compare the medications from the external list to the current list
- Clarify inconsistencies/discrepancies (e.g., omissions, duplications, contraindications, unclear information, and changes)
- Develop a list of medications to be prescribed at discharge or transfer
- Communicate the new list to the patient and/or appropriate caregiver(s)
- Ensure that patient and/or caregiver(s) understand the medication information upon discharge

Specific medication issues identified as being problematic include missing medication information from transfer orders, lack of information on medications provided in the acute setting, incomplete medication records, discrepancies between hospital regimen and discharge summary, and missing information pertaining to the patient's tolerance of a medication regimen. Although medication reconciliation is an important aspect in patient safety, there is a lack of consensus and evidence about the best effective methods of implementing this process. Physician-led and electronic medication reconciliation in hospitals are effective strategies to reduce medication discrepancies, however, the impact of these interventions is uncertain due to the low quality of evidence (Choi and Kim, J Clin Pharm Ther 2019, 44: 932-45; Redmond et al., Cochrane Database Syst Rev 2018, 8: CD010791). Trained pharmacy technicians, under the direction of a licensed pharmacist, may be an option for developing and expanding medication reconciliation processes (Irwin et al., Hosp Pharm 2017, 52: 44-53).

57:

Outpatient management contraindicated refers to the following:

- The patient whose medical condition is so fragile that going to the clinic or medical practitioner's office for treatment would put the patient at risk
- The patient with extensive equipment (e.g., ventilators, specialized beds) where travel outside the home would be cumbersome and places the patient at risk
- The treatment goal is to adapt the home environment for increased independence (e.g., an individual who is blind who needs mobility education)

58:

Outpatient management unavailable refers to the following:

- The patient whose treatment (e.g., dressing changes) is so extensive that it takes too long or is too frequent (e.g., twice daily) to be done in an outpatient clinic or medical practitioner's office
- The scheduling of the treatment falls outside the medical practitioner's office or clinic hours
- Travel distance to receive care is excessive (e.g., greater than 1 hour)
- The service is unavailable in the outpatient setting

59:

Skilled nursing services refer to services that must be provided by a licensed nurse qualified to assess and monitor patient condition(s) and provide medical treatment and/or teaching to patients who have skilled nursing needs. Examples of Home Care level of care interventions include skilled assessment, intervention, or education for any of the following:

- New treatment or medication regimen
- Adjustment in the treatment or medication regimen
- Infusion therapy administration, teaching, or access device management
- Management and evaluation of a care plan for patients with an active comorbidity and multiple care needs
- Chronic disease requiring a disease management program (e.g., asthma, heart failure, chronic obstructive pulmonary disease, diabetes)
- End-stage disease, hospice, or palliative care

60:

Skilled nursing interventions refer to services that are provided or supervised by a qualified or licensed professional when the care needs of the patient require assessment, observation, monitoring, and/or education to achieve the medically desired result. Close observation and assessment by nursing may be necessary at this level of care. This includes nursing care such as medication administration (excludes routine), vital signs monitoring, and/or other interventions not listed.

61:

The Home Care criteria are used for a patient who has been admitted or is expected to be admitted for home care. When criteria are met for this level of care and it is not available in your area, we recommend you use the next higher level of care.

Home care can include specialty services provided by a psychiatric nurse following a specialized treatment plan developed by a psychiatrist or advanced practice registered nurse. It is intended for treatment and monitoring of a patient's psychiatric condition that prevents the patient from leaving the house or makes leaving the house extremely difficult or unsafe (Centers for Medicare & Medicaid Services. Chapter 7, In: Medicare benefit policy manual. 2022). Home care can also be used for a patient with severe and persistent mental illness with a history of nonadherence to medication.

Evaluation and Treatment

Programming may differ based upon legislative and geographical variances and is subject to organizational policy; however, at a minimum, it should include:

- Care coordination with treating physician and other providers
- Clinical assessment every visit
- Education every visit
- Medication assessment, teaching, and management every visit
- Medication reconciliation initiated within first visit
- Psychosocial assessment within first visit
- Substance evaluation within first 2 visits

For those patients who are covered under a Medicare benefit, the Affordable Care Act mandates that a certifying physician or allowed practitioner have a face-to-face visit with the patient within 90 days prior to the start of care, or within 30 days after the start of care. There must be documentation of the patient's condition that supports the need for home care services and the requirement of homebound status (Centers for Medicare & Medicaid Services. Chapter 7, In: Medicare benefit policy manual. 2022).

62:

Support system includes friends, family, social services, health care delivery providers or agencies, sponsors, clergy, and self-help groups.

63:

The initial clinical assessment is performed by a licensed professional and includes a comprehensive review of the patient's presenting diagnosis and a review of body systems. Identification of current and potential medical needs and health problems can be identified by the clinical assessment. The clinical assessment includes:

- History and physical exam (e.g., vital signs, height, weight)
- Pain assessment includes cause, intensity, quality, onset, duration, and effects on quality of life (e.g., activities of daily living (ADLs), instrumental activities of daily living (IADLs), and interpersonal relationships). Pain relievers should also be included
- Nutritional and hydration status
- Functional ability
- Safety and infection control measures
- Prescribed and over-the-counter (OTC) medications, herbal supplements, and home remedies
- Patient's and/or caregiver's understanding of medication use, dosing, side effects, and adherence to the medication or treatment regimen
- Patient's and/or caregiver's understanding of the illness or disease process and potential long-term complications
- Patient's and/or caregiver's coping strategies to deal with the illness or injury and ability to follow the plan of care

64:

For specific level of care recommendations for the treatment of psychiatric and substance use disorders, refer to the InterQual® Adult and Geriatric Psychiatry, InterQual® Child and Adolescent Psychiatry, or InterQual® Substance Use Disorders products.

65:

Goal-directed therapy includes one or more of the following:

- Activities of daily living (ADLs) (e.g., feeding, dressing, bathing, grooming, toileting, functional mobility)
- Bed mobility or transfers (e.g., transfers to chair, toilet, commode, tub, shower, car)
- Cognitive, speech, language, or swallowing retraining
- Home and lifestyle modification (e.g., assessment and training focused on the home, job, school, or work environment)
- Pain management techniques (e.g., assessment and interventions used to relieve pain or decrease its intensity)
- Pulmonary rehabilitation (e.g., energy conservation, speech, breathing re-education, postural drainage, percussion, or vibration)
- Positioning and splinting
- Range of motion and strengthening
- Wheelchair mobility
- Ambulation

66:

Rehabilitation potential refers to the probability that therapy and medical goals are realistic and attainable based on patient's prior level of function, severity of illness or injury, and the extent of impairments.

67:

Functional assistance levels are based upon the patient's function during tasks and activities necessary to return to household mobility or ambulation. The functional assistance level required for each individual task or activity (e.g., mobility, activities of daily living (ADLs)) may vary.

The following terms are commonly used in the post-acute setting:

- Independent - Patient can safely and within a reasonable amount of time perform a task (or developmentally appropriate task) without physical or cognitive assistance or supervision

- Modified Independent - Patient performs an activity with a supportive device, adaptive equipment, and/or prosthetic or orthotic device. Additional time may be required to complete the activity and/or there are safety (risk) considerations
- Supervision - Patient performs an activity with standby or distant supervision or setup. Verbal cueing or coaxing, without physical contact or setup of items and application of orthoses may be required when patient's safety awareness is impaired
- Minimum or limited Assistance - Patient performs at least 75%(0.75) of an activity and requires some physical contact to steady, guide, or move
- Moderate or extensive Assistance - Patient performs at least 50%(0.50) of an activity and requires physical assistance for functional mobility or ADLs
- Maximum Assistance - Patient performs 25%(0.25) to 50%(0.50) of an activity and requires physical assistance for functional mobility or ADLs
- Total Assistance or dependence - Patient performs less than 25%(0.25) of an activity and may require total assistance for functional mobility or ADLs

68:

The intensity of rehabilitation therapies is a driver for admission to an acute inpatient rehabilitation facility. Documentation should reflect the need for required and ongoing intensive therapies. The determination of whether a patient requires and can tolerate an intensive rehabilitation program of at least 15 hours is not based on any one single piece of documentation. All clinical documentation should be evaluated by the reviewer to determine whether the patient is demonstrating the need for services and has the ability to tolerate them. Patients who do not require this level of intensive therapy should be treated at a lower level of care. Rehabilitation therapies should be reasonable and medically appropriate and should not exceed a patient's level of tolerance. If there are clinical features which preclude the patient's ability to tolerate the intensity threshold of at least 15 hours per week, secondary review would be appropriate (Centers for Medicare & Medicaid Services, Medicare Benefit Policy Manual Chapter 1 - Inpatient Hospital Services Covered Under Part A. 2021; Miller and Forer, Pm r 2021, 13: 535-9; Forrest et al., Medicine (Baltimore) 2019, 98: e17096).

69:

Comorbid conditions include new onset or worsening behavioral symptoms, heart failure, deep vein thrombosis, uncontrolled diabetes, immunocompromised states, malignant or end-stage disease, malnutrition, renal insufficiency or end-stage renal disease, infection, ventilator dependence, noninvasive positive pressure ventilation or respiratory insufficiency, complex wound care, and new functional impairment(s) with limitation.

70:

Ventilation includes invasive ventilation and noninvasive positive pressure ventilation (NIPPV), including continuous positive airway pressure (CPAP) and bilevel positive airway pressure (BiPAP).

71:

Notable causes for prolonged mechanical ventilation include:

- Ventilator-acquired pneumonia
- Renal failure
- Heart failure
- Exacerbation of chronic obstructive pulmonary disease (COPD)
- Chest wall disorder
- Neuromuscular diseases
- Traumatic brain injury
- Sepsis
- Complications following cardiac surgical procedures

Major contributing factors for failure to wean include (Ambrosino and Vitacca, Multidiscip Respir Med 2018, 13: 6; Davies et al., Br J Anaesth 2017, 118: 563-9):

- Increased work of breathing
- Impaired respiratory drive
- Inspiratory muscle weakness

72:

Long-Term Acute Care (LTAC) is a recognized level of care designation by the Centers for Medicare and Medicaid Services (CMS) for acute care hospitals. It is defined by federal statutes as a level of care for patients requiring an average length of stay of 25 days or more (Centers for and Medicaid Services, Fed Regist 2018, 83: 41144-784). Patients who require a short stay for treatment may be more appropriate for the subacute or skilled nursing facility (SNF) level of care.